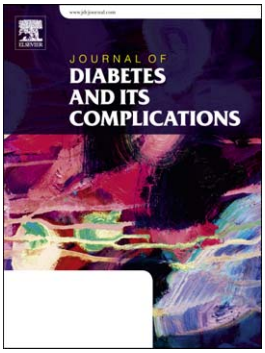


Accepted Manuscript

Bile Acid Sequestrants for Glycemic Control in Patients with Type 2 Diabetes: A Systematic Review with Meta-Analysis of Randomized Controlled Trials

Morten Hansen, David P. Sonne, Kristian H. Mikkelsen, Lise Lotte Gluud, Tina Vilsbøll, Filip K. Knop

PII: S1056-8727(16)30649-3
DOI: doi: 10.1016/j.jdiacomp.2017.01.011
Reference: JDC 6942

To appear in: *Journal of Diabetes and Its Complications*

Received date: 18 October 2016
Revised date: 18 January 2017
Accepted date: 19 January 2017

Please cite this article as: Hansen, M., Sonne, D.P., Mikkelsen, K.H., Gluud, L.L., Vilsbøll, T. & Knop, F.K., Bile Acid Sequestrants for Glycemic Control in Patients with Type 2 Diabetes: A Systematic Review with Meta-Analysis of Randomized Controlled Trials, *Journal of Diabetes and Its Complications* (2017), doi: 10.1016/j.jdiacomp.2017.01.011

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

ACCEPTED MANUSCRIPT

Bile Acid Sequestrants for Glycemic Control in Patients with Type 2 Diabetes: A Systematic Review with Meta-Analysis of Randomized Controlled Trials

Morten Hansen[1,2*], David P. Sonne[1] Kristian H. Mikkelsen[1,2],
Lise Lotte Gluud[1,3], Tina Vilsbøll[1] and Filip K. Knop[1,2]

[1]Center for Diabetes Research, Gentofte Hospital, University of Copenhagen, Hellerup, Denmark; [2]NNF Center for Basic Metabolic Research and Department of Biomedical Sciences, Faculty of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark; [3]Gastro Unit, Medical Division, Hvidovre Hospital, University of Copenhagen, Hvidovre, Denmark; *Current affiliation: Novo Nordisk A/S, Bagsværd, Denmark

Word count (excluding title page, abstract and references): 5,250
Protocol registration: PROSPERO CRD42012002552

Correspondence to:
Prof. Filip K. Knop, MD PhD
Center for Diabetes Research
Gentofte Hospital, University of Copenhagen
Kildegårdsvej 28
DK-2900 Hellerup
Denmark
Tel: +45 3867 4266
Fax: +45 3867 7661
E-mail: filipknop@dadlnet.dk

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

ABSTRACT

Aim To evaluate the effects of bile acid sequestrants (BASs) versus placebo, no intervention or active comparators on glycemic control in type 2 diabetes.

Methods Data were retrieved and a systematic review with meta-analyses was performed. We evaluated bias control and subgroup and sensitivity analyses were performed to evaluate heterogeneity and bias.

Results We included 17 trials with a total of 2,950 patients randomized to BASs (colesevelam or colestimide) versus placebo, no intervention, statins or sitagliptin. Random-effects meta-analysis showed that patients randomized to BASs had a lower hemoglobin A_{1c} at the *end of treatment* compared with the control group (mean difference -0.55%; 95% confidence interval -0.64 to -0.46). Analysis of trials with low risk of bias in all domains confirmed the findings. Data on adverse events were limited. There were no differences between trials stratified by the control group and no evidence of publication bias or small study effects.

Conclusions Our analyses found that BAS treatment improves glycemic control. The size of the effect was clinically relevant and despite limited safety data, our findings support the inclusion of BASs in current diabetes management algorithm**s** for type 2 diabetes.

Key words Bile acid sequestrants; colesevelam; colestimide; meta-analysis; glycemic control; type 2 diabetes

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

1. INTRODUCTION

Type 2 diabetes is associated with obesity, hypertension, dyslipidemia and an increased risk of cardiovascular disease. Glycemic control is monitored by glycated hemoglobin A_{1c} (HbA_{1c}), which, in addition to reflecting overall plasma glucose concentration, is an independent predictor of cardiovascular disease (Gall, Borch-Johnsen, Hougaard, Nielsen, & Parving, 1995; Stratton et al., 2000). However, the benefit of intensive glycemic control on macrovascular complications is controversial and individualized glycemic targets are recommended as the best way to manage the disease (ADVANCE Collaborative Group et al., 2008; Inzucchi et al., 2015). Dyslipidemia exacerbates the risk of cardiovascular complications associated with type 2 diabetes and treatment with statins reduce cardiovascular mortality in type 2 diabetes (Gaede et al., 2003). Numerous antidiabetic drugs and lipid-lowering drugs are used in the management of type 2 diabetes, but not all only few patients reach the treatment goals (Cheung et al., 2009; Inzucchi et al., 2015).

Bile acid sequestrants (BASs) constitute a class of drugs that have been used for cholesterol-lowering treatment in decades (Lyons, Webster, Fowler, & Petrie, 1994; Zieve, Kalin, Schwartz, Jones, & Bailey, 2007). The BASs comprise first-generation cholestyramine and colestipol and second-generation colesevelam hydrochloride (referred to as colesevelam) and colestimide (also known as colestilan, marketed only in Japan). BASs form a nonabsorbable complex with bile acids in the intestine, which prevents reabsorption and increases fecal excretion of bile acids (Rosenbaum, Petersen, Ducharme, Markham, & Goldberg, 1997). Consequently, bile acid synthesis is increased, which diverts circulating low-density lipoprotein (LDL) cholesterol into bile acids thereby reducing plasma LDL cholesterol. Cholestyramine was the first lipid-altering drug to show, in a randomized trial, a reduced risk of myocardial infarction and death by coronary heart disease.("The Lipid Research Clinics

ACCEPTED MANUSCRIPT

Coronary Primary Prevention Trial results. II," 1984; "The Lipid Research Clinics Coronary Primary Prevention Trial results. I," 1984). In 1994, Garg and Grundy described that cholestyramine may also reduce fasting plasma glucose in patients with type 2 diabetes (Garg & Grundy, 1994). Fourteens years later, in 2008, colesevelam was approved by the US Food and Drug Administration (FDA) for the treatment of hyperglycemia in type 2 diabetes and colesevelam is now included in diabetes management algorithms (Garber et al., 2013; Inzucchi et al., 2015).

Published data suggest that BASs improve glycemic control. However, not all trials report this outcome in a consistent manner, which may suggest reporting bias. In order to describe the effects of BAS treatment in patients with type 2 diabetes we gathered published and unpublished data and performed a systematic review with meta-analyses, subgroup and sensitivity analyses.

2. MATERIALS AND METHODS

This systematic review was based on a registered (PROSPERO CRD42012002552) and published protocol (Hansen et al., 2012) prepared and reported based on previous recommendations ("Cochrane Handbook for Systematic Reviews of Interventions"; Moher, Liberati, Tetzlaff, & Altman, 2009). Our main objective was to evaluate the effects of BASs versus placebo or no interventions on glycemic control and adverse events in type 2 diabetes. We originally planned to compare BAS versus other antidiabetic interventions, but only identified one trial comparing BASs versus rosiglitazone or sitagliptin (Rigby et al., 2010). We decided not to include data from the rosiglitazone arm in our analyses because the drug has been associated with increased risk of myocardial infarction but not cardiovascular or all-cause mortality (Nissen & Wolski, 2010). Our primary endpoint was glycemic control

ACCEPTED MANUSCRIPT

(HbA_{1c}). Secondary endpoints included adverse events and changes in fasting plasma glucose, total cholesterol, high-density lipoprotein (HDL) cholesterol, LDL cholesterol, triglycerides and body weight.

2.1 Data Sources and Searches

We included randomized controlled trials, irrespective of date, publication status, language, bias control and predefined outcome. Eligible trials were identified by electronic searches in The Cochrane Library, Medline, Embase, the Web of Science and clinicaltrials.gov. The last search was made on 4 August 2016. The search terms in the Cochrane Library were ("bile acid sequestrant" or "cholestyramine" or "colestimide" or "colestipol" or "colesevelam") and "type 2 diabetes". The search strategy was adjusted to fit the other databases and combined with the Cochrane Search String for the identification of randomized clinical trials ("Cochrane Handbook for Systematic Reviews of Interventions"). We also searched the WHO International Clinical Trial Registry Platform and conference proceedings from the annual scientific meetings of the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). The reference lists of relevant articles were manually examined and relevant trials identified. Additional information on trials was sought through correspondence with authors and pharmaceutical companies.

2.2 Study Selection

Two authors independently reviewed the trial inclusion and exclusion criteria. Excluded trials were listed with the reason for exclusion. We included trials of adult patients of both genders with type 2 diabetes as defined by the authors of the included trials. The BAS groups included cholestyramine, colesevelam colestimide or colestipol, irrespective of daily dose or duration of the trial. The control groups were placebo, no intervention or other active comparators. Co-interventions were allowed if administered equally to the intervention groups.

ACCEPTED MANUSCRIPT

2.3 Data extraction and Quality Assessment

Data extraction was performed independently by three authors based on standardized extraction forms. Disagreements were resolved through discussion. Corresponding authors were contacted if the required information was not described in the published reports. The risk of bias was assessed by separate domains: random sequence generation, allocation concealment, blinding of participants, personnel and outcome assessment, incomplete outcome data, selective reporting and other bias ("Cochrane Handbook for Systematic Reviews of Interventions").

2.4 Data Synthesis and Analysis

Analyses were performed using STATA version 13 (Stata Corp, College Station, TX, USA). The meta-analyses were performed using random-effects based on expected clinical heterogeneity. We repeated the analyses using fixed-effect models. In all analyses, the overall conclusion of the two models concurred, but random-effects models provided the most conservative estimate of the intervention effect. We therefore only report the result of the random-effects meta-analyses. Results of the meta-analyses were expressed as mean differences (MD) for continuous outcomes and relative risks (RR) for dichotomous outcomes, both with 95% confidence intervals (CI). We used I^2 values as markers of heterogeneity with the following thresholds: 0% to 40%: not important; 30% to 60%: moderate heterogeneity; 50% to 90%: substantial heterogeneity; 75% to 100%: considerable heterogeneity (Higgins & Thompson, 2002). Funnel plots and regression tests for funnel plot asymmetry in meta-analysis were performed for meta-analyses with at least ten trials to assess small study effects (Egger test for continuous and Harbord for dichotomous outcomes). The effects of interventions were examined using subgroup and sensitivity analyses. These analyses were performed according to type of BAS, type of control, risk of bias, type of data (published or

ACCEPTED MANUSCRIPT

unpublished) and country of origin. We performed a post-hoc subgroup analysis on the country of origin to evaluate the potential influence of ethnicity. Differences between subgroups were expressed as P values (test for subgroup differences).

3. RESULTS

3.1 Search Results and Trial Characteristics

The initial search generated 383 references (**Fig. 1**) of which 20 fulfilled the predefined inclusion criteria. We had to exclude three trials from our quantitative analyses because we were unable to obtain the required data (Garg & Grundy, 1994; Mazze et al., 2013; Yamakawa, Takano, Utsunomiya, Kadonosono, & Okamura, 2007). Thus, 17 randomized clinical trials were included in our analyses. The trials were published as full paper articles between 2007 and 2014 (**Table 1**). We retrieved unpublished information about trial design and outcome measures from 16 trials (Bays, Goldberg, Truitt, & Jones, 2008; Fonseca, Rosenstock, Wang, Truitt, & Jones, 2008; Goldberg, Fonseca, Truitt, & Jones, 2008; Henry et al., 2012; Kato et al., 2011; Kondo & Kadowaki, 2010; Neda et al., 2012; Rigby et al., 2010; J. Rosenstock, Truitt, et al., 2014; J. Rosenstock, Rigby, Ford, Tao, & Chou, 2014; Julio Rosenstock et al., 2010; Schwartz et al., 2010; Smushkin et al., 2013; Suzuki et al., 2012; Takebayashi, Suetsugu, Matsumoto, Aso, & Inukai, 2009; Zieve et al., 2007). We included 12 colesevelam trials and 5 colestimide trials. In total, 2,950 patients were randomized to colesevelam or colestimide versus placebo/no intervention (14 trials),(Bays et al., 2008; Beysen et al., 2012; Fonseca et al., 2008; Goldberg et al., 2008; Henry et al., 2012; Kondo & Kadowaki, 2010; Neda et al., 2012; J. Rosenstock, Truitt, et al., 2014; J. Rosenstock, Rigby, et al., 2014; Julio Rosenstock et al., 2010; Schwartz et al., 2010; Smushkin et al., 2013; Suzuki et al., 2012; Zieve et al., 2007) statins (2 trials),(Kato et al., 2011; Takebayashi et al.,

ACCEPTED MANUSCRIPT

2009) or sitagliptin (1 trial).(Rigby et al., 2010) The daily dose was 3.75 g for colesevelam and 3 to 4.5 g for colestimide. The duration of treatment was 8 to 26 weeks. Both drugs were administered as tablets. The collateral antidiabetic interventions (administered to both intervention groups) included diet alone, alpha-glucosidase inhibitors, meglitinides, metformin, sulfonylurea, thiazolidinediones, insulin and insulin analogues. Two trials included patients not previously treated with antidiabetic medication (J. Rosenstock, Rigby, et al., 2014; Julio Rosenstock et al., 2010).

The patient inclusion and exclusion criteria were comparable across the included trials. The colesevelam trials were conducted in the USA, Colombia, Mexico and India. One of the trials was a single center study (Schwartz et al., 2010). The colestimide trials were conducted in Japan and only one of these trials was a multicenter study (Kondo & Kadowaki, 2010). The patients in the colesevelam trials were African-American, Asian, Caucasian or Hispanic. Patients in the colestimide trials were Japanese. The mean HbA_{1c} at baseline ranged from 6.7% to 8.7% (50.8 mmol/mol to 71.6 mmol/mol), mean fasting plasma glucose from 7.0 mmol/L to 10.3 mmol/L, and mean BMI ranged from 25.1 kg/m^2 to 34.9 kg/m^2 (**Table 2)**.

3.2 Bias Control

Based on the published information and information retrieved from the corresponding authors, the allocation sequence was generated by a computer or table of random numbers in 15 trials. Information about allocation sequence generation could not be retrieved in the remaining 2 trials (Kato et al., 2011; Suzuki et al., 2012). Accordingly, we classed the allocation sequence as “low risk of bias” in 15 trials and “uncertain risk of bias” in 2 trials (Kato et al., 2011; Suzuki et al., 2012). In 16 trials, the allocation was concealed using central randomization, sealed serially numbered opaque sealed envelopes or administration of identically appearing coded drug bottles/containers. The allocation concealment was classed

ACCEPTED MANUSCRIPT

as “low risk of bias” in these trials and “uncertain risk of bias” in the remaining trial (Suzuki et al., 2012). Twelve trials that were double-blinded using a placebo (blinding of participants, personnel and outcome assessors) were classed as low risk of performance and detection bias. The remaining five trials were open and classed as high risk of performance and detection bias (Kato et al., 2011; Neda et al., 2012; Rigby et al., 2010; Suzuki et al., 2012; Takebayashi et al., 2009). All patients were accounted for in 16 trials, which were classed as low risk of attrition bias (incomplete outcome data). One trial did not account for patients with incomplete outcome data (Kondo & Kadowaki, 2010). All 17 trials defined and reported clinically relevant outcome measures, and we did not identify differences between the protocoled outcome measures (published in e.g. trial registries) and reported outcomes measures. Accordingly, all trials were classed as low risk of reporting bias. No other biases were identified (**Table 3**).

3.3 Hemoglobin A1c

We were able to gather data on glycemic control based on the *end of treatment* HbA_{1c} from 10 trials with a total of 1,369 patients (**Fig. 2A**). Random-effects meta-analysis showed that patients randomized to BASs had a lower *end of treatment* HbA_{1c} compared with the control group (MD -0.55%; -0.64 to -0.46; I^2=13%). Regression analyses showed a trend towards bias or other small study effects (P=0.06). Subgroup analyses showed no difference (P=0.15) between colesevelam (MD -0.53%, -0.65 to -0.42) and colestimide trials (MD -0.64%; -0.79 to -0.48). Likewise, there was no difference (P=0.31) between trials with placebo (MD -0.51%; -0.59 to -0.42), statin (MD -0.74; -1.139 to -0.335) or no intervention control groups (MD -0.62%; -0.78 to -0.45). Sensitivity analyses of trials with a low risk of bias, assessed using individual domains, confirmed the findings (MD -0.53%; -0.66 to -0.40; I^2=39.9%; 5

ACCEPTED MANUSCRIPT

trials). Post-hoc analyses based on the country of origin of the trials did not show any difference between trials conducted in Japan and the USA (data not shown).

Eleven trials including a total of 2,630 patients reported the *mean change* in HbA_{1c}. Random-effects meta-analysis showed that patients randomized to BASs had a lower HbA_{1c} compared with controls (MD -0.40%; -0.49 to -0.30; I^2=96.9%). No evidence of small study effects was identified (P=0.70). Subgroup analyses showed a difference (P<0.001) between trials on colesevelam (MD -0.29%; -0.30 to -0.27; I^2=96.7%) and the trial on colestimide (MD -0.88%; -1.04 to -0.72). Sensitivity analysis of the individual control groups showed a different effect of BAS treatment in trials with a placebo control group (MD -0.30%; -0.31 to -0.29) and the trial with sitagliptin (MD 0.10%; 0.05 to 0.15). Again, the heterogeneity between the trials with a placebo control group was considerable (I^2=90.6). Sensitivity analyses on trials with a low risk of bias assessed using individual domains, confirmed the findings (MD -0.44%; -0.51 to -0.37; I^2=91.6%; 9 trials). No effect of publication status was observed between the trials analyzed. Post-hoc analyses based on the country of trial origin showed a greater reduction in HbA_{1c} in the trials that were conducted in Japan versus the trials conducted in USA (data not shown). Again, heterogeneity within in the colesevelam sub-group was high (I^2=96.7%), and the test likely to be invalid.

3.4 Fasting Plasma Glucose

Ten trials with a total of 1,369 patients reported *end of trial* fasting plasma glucose and random-effects meta-analysis showed a beneficial effect of BAS treatment versus control (MD -1.07 mmol/L; -1.22 to -0.91; I^2=9.8%) (**Fig. 2B**). Forest plot or regression analyses of forest plot asymmetry showed no evidence of small study effects (P=0.76). Subgroup analyses showed a significant difference (P=0.013) between trials on colesevelam (MD -0.89 mmol/L; -1.09 to -0.69) or colestimide (MD -1.21 mmol/L; -1.37 to -1.06). There was a statistically

ACCEPTED MANUSCRIPT

significant greater reduction in *end of trial* fasting plasma glucose in trials with no intervention control group (P=0.04) versus trials with a placebo control group or a statin treatment control group. Sensitivity analyses on trials with a low risk of bias, assessed using individual domains, confirmed the findings (MD -0.87 mmol/L; -1.08 to -0.66; I^2=0.0%; 5 trials). No effect of publication status was observed between the trials analyzed. Post-hoc analyses based on country of trial origin showed a greater reduction in fasting plasma glucose in the trials that were conducted in Japan versus the trials conducted in USA (data not shown). However, the majority of the trials conducted in Japan were unblinded. We gathered data on *mean change* in fasting plasma glucose from 10 trials including a total of 2,531 patients. Random-effects meta-analysis showed that BAS treatment reduced fasting plasma glucose compared to control (MD -0.59 mmol/L; -0.61 to -0.57; I^2=98.8%) (data not shown). Given the considerable heterogeneity, this result should be interpreted cautiously. Forest plot or regression analyses of forest plot asymmetry showed no evidence of small study effects (P=0.98). Subgroup analyses showed a significant difference (P<0.01) between trials on colesevelam (MD -0.59 mmol/L; -0.60 to -0.57) and colestimide -1.30 mmol/L; -1.64 to -0.96). However, the heterogeneity within in the colesevelam sub-group was high (I^2=98.9%), and the test likely to be invalid. Sensitivity analysis including the trial with a sitagliptin comparison group did not change the overall conclusion, but increased between-trial heterogeneity (MD 0.22 mmol/L; 0.11 to 0.33). Sensitivity analyses on trials with a low risk of bias assessed using individual domains confirmed the findings (MD -0.80 mmol/L; -1.04 to -0.56; I^2=98.7%; 8 trials). No effect of publication status was observed between the trials analyzed. Post-hoc analyses based on the country of trial origin showed a greater reduction in fasting plasma glucose in the trials that were conducted in Japan versus the trials conducted in USA (data not shown).

ACCEPTED MANUSCRIPT

3.5 Total Cholesterol

Six trials (543 patients) reported *end of trial* total cholesterol. Random-effects meta-analysis showed no difference between BAS treatment versus the control group (MD -0.12 mmol/L; -0.56 to 0.34; I^2=93.1%). There was evidence of small study effects (P=0.02). Subgroup analyses showed a significant difference (P<0.001) between colesevelam (MD -0.18 mmol/L; -0.37 to 0.01) and colestimide trials (MD -0.69 mmol/L; -0.78 to -0.60). Heterogeneity within the colestimide sub-group was considerable (I^2=95.9%), and the test likely to be invalid. There was a statistically significant greater reduction in total cholesterol in trials with no intervention control group (P<0.001) versus trials with a placebo control group, whereas statin treatment resulted in a greater reduction in total cholesterol versus BAS treatment (P<0.001). Sensitivity analyses on trials with a low risk of bias assessed using individual domains confirmed the findings (MD -0.21 mmol/L; -0.46 to 0.05; I^2=23.2%; 2 trials). No effect of publication status was observed between the trials analyzed. One trial (172 patients) reported *mean change* in total cholesterol, and found a significant reduction after colestimide treatment versus placebo (MD -0.70 mmol/L; -0.87 to -0.53) (low risk of bias in individual domains).

3.6 LDL Cholesterol

Random-effects meta-analysis of the 6 trials (548 patients) reporting *end of trial* LDL cholesterol after BAS compared to control treatment showed no difference between the interventions (MD 0.12 mmol/L; -0.50 to 0.74; I^2=95.7%). There was no evidence of small study effects (P=0.60). Subgroup analyses showed a significant difference (P<0.001) between colesevelam (MD -0.33 mmol/L; -0.49 to -0.17) and colestimide trials (MD 0.57 mmol/L; 0.37 to 0.77). Heterogeneity within in the colestimide subgroup was considerable (I^2=97.0%), indicating that the test is likely to be invalid. There was a statistically significant greater reduction in LDL cholesterol after BAS treatment in trials with placebo (MD -0.33 mmol/L; -

ACCEPTED MANUSCRIPT

0.49 to -0.17) and no intervention control group (MD -0.44 mmol/L; -0.75 to -0.13) (P<0.01) whereas statin treatment resulted in a greater reduction in LDL cholesterol versus BAS treatment (MD 1.23 mmol/L; 0.98 to 1.48) (P<0.001). Sensitivity analyses of trials with a low risk of bias assessed using individual domains confirmed the findings (MD -0.33; -0.55 to -0.12; I^2=22.2%; 2 trials). No effect of publication status was observed between the trials analyzed. Two trials (458 patients) reported *mean change* in LDL cholesterol after treatment with BAS versus placebo. Random-effects meta-analysis showed a beneficial effect of BAS treatment (MD -0.61 mmol/L; -0.95 to -0.26; I^2=83.8%) (low risk of bias in individual domains in both trials).

3.7 HDL Cholesterol

Six trials (548 patients) reported *end of trial* HDL cholesterol. Random-effects meta-analysis showed no difference between BAS treatment and control (MD -0.02 mmol/L; -0.11 to 0.07; I^2=34.2%). There was no evidence of small study effects (P=0.48). Subgroup analyses showed no significant difference (P=0.72) between colesevelam (MD 0.01 mmol/L; -0.09 to 0.10) and colestimide trials (MD -0.04 mmol/L; -0.14 to 0.07) and there was no difference between trials with placebo, no intervention and statin control group (P=0.18). Sensitivity analyses of trials with a low risk of bias assessed using individual domains confirmed the findings (MD -0.13 mmol/L; -0.30 to 0.04; I^2=0.0%; 2 trials). No effect of publication status was observed between the trials analyzed.

3.8 Triglycerides

Random-effects meta-analysis of the 7 trials (591 patients) reporting *end of trial* serum triglycerides, showed a trend towards increase in triglycerides after BAS treatment versus control (MD 0.23 mmol/L; -0.04 to 0.49; I^2=65.5%). Evidence of small study effects was identified (P=0.03). Subgroup analyses showed a difference (P<0.025) between colesevelam

ACCEPTED MANUSCRIPT

(MD 0.27 mmol/L; 0.02 to 0.53) and colestimide trials (MD -0.05 mmol/L; -0.15 to 0.06). However, heterogeneity within in the colestimide subgroup was high (I^2=68.9%), and the test is likely to be invalid. There was no difference between trials with placebo (MD 0.27 mmol/L; 0.02 to 0.53) no intervention (-0.07 mmol/L; -0.18 to 0.04) and statin control group (0.25 mmol/L; -0.14 to 0.63) (P>0.03). Sensitivity analyses of trials with a low risk of bias assessed using individual domains confirmed the findings (MD 0.35 mmol/L; -0.15 to 0.84; I^2=63.0%; 2 trials). No effect of publication status was observed between the trials analyzed.

Only 1 trial (172 patients) reported *mean change* in serum triglycerides and showed no treatment difference after colestimide treatment compared with placebo (MD 0.14 mmol/L; -0.22 to 0.50) (low risk of bias in individual domains). The large-scale trials comparing colesevelam versus placebo did not report raw outcome data on serum triglycerides. Instead, they reported median change. Three trials (553 patients) reported non-significant increases in serum triglyceride between 4.7% and 12.2% (2 of these trial had low risk of bias in all domains), and 5 trials (1,709 patients) reported significant increases in serum triglyceride of 14.1% and 21.5% compared to control group (low risk of bias in individual domains in all trials). These data were not eligible for meta-analysis.

3.9 Body Weight

Six trials (1,156 patients) reported *mean change* in body weight after BAS treatment versus placebo. Random-effects meta-analysis showed no treatment difference (MD -0.02 kg; -0.27 to 0.24; I^2=0.0%) (**Fig. 3, online supplementary**). There was no evidence of small study effects (P=0.177). Subgroup analyses did not show a difference (P=0.33) between trials on colesevelam (MD 0.08 kg; -0.24 to 0.41) or colestimide (MD -0.18 kg; -0.60 to 0.24). All 6 trials included a placebo control group. Sensitivity analyses on trials with a low risk of bias

ACCEPTED MANUSCRIPT

assessed using individual domains confirmed the findings (MD -0.03 kg; -0.29 to 0.24; I^2=0.0%; 5 trials). No effect of publication status was observed between the trials analyzed.

3.10 Adverse Events

Thirteen trials (2,714 patients) reported adverse events defined as "any adverse event". There were more adverse events in the group treated with BASs versus the control group (RR 1.09 (95% CI: 1.02 to 1.17; P=0.01). Heterogeneity was absent (I^2=0.0%). Sensitivity analyses on trials with a low risk of bias assessed using individual domains confirmed the findings (data not shown).

Significant more serious adverse events were observed in the group treated with BASs versus the control group (RR 1.56; 0.01 to 2.40; P=0.04; 9 trials; 2,484 patients), and heterogeneity was absent (I^2=0.0%). The number of adverse events deemed drug-related by the trial investigators was significantly higher in the group treated with BASs versus the control group (RR 1.62; 1.20 to 2.19; P=0.008; 13 trials; 2,655 patients). Heterogeneity was substantial (I^2=55.5%). The number of drug-related withdrawals did not differ significantly between the group treated with BASs versus the control group (RR 1.48; 0.91 to 2.42; P=0.116; 9 trials; 2,501 patients). Heterogeneity was absent (I^2=0.0%). Eight trials did not have any drug-related withdrawals. The most frequent adverse events were of gastrointestinal origin, constipation being the most prevalent (RR 3.13; 1.88 to 5.21; P<0.001; 1,537 patients). Heterogeneity was absent (I^2=0.0%). No effect of publication status was observed between the trials analyzed.

4. DISCUSSION

This systematic review with meta-analyses includes data from randomized clinical trials evaluating the glucose-lowering effect of the BASs colesevelam and colestimide in patients

ACCEPTED MANUSCRIPT

with type 2 diabetes. Our key finding is that colesevelam and colestimide induced clinical relevant reductions in HbA_{1c} by 0.40% to 0.55% (4.4 mmol/mol to 6.0 mmol/mol) and fasting plasma glucose by 0.65 mmol/L to 1.07 mmol/L compared to the control group. Also, we confirm that colesevelam and colestimide reduce LDL cholesterol, although to a lesser extent than statin treatment, and increases the risk of constipation. These findings were confirmed by analysis of the trials with low risk of bias in all domains.

Some of the identified trials reported postprandial glucose concentrations obtained after different interventions such as 75-g oral glucose tolerance test (OGTT) (Brufau et al., 2010; Henry et al., 2012; Kato et al., 2011), varying types of mixed meal tolerance tests (Beysen et al., 2012; Marina et al., 2012; Schwartz et al., 2010; Smushkin et al., 2013; Zieve et al., 2007) and in one trial the intervention was not reported (Kondo & Kadowaki, 2010). While postprandial glucose was not a pre-specified endpoint in the study protocol and the results were obtained under different conditions precluding meaningful meta-analysis, the results did provide insights into the potential mode of action of BAS. With the exception of the trial by Kato *et al*., no difference was observed in postabsorptive glucose concentrations between BAS and placebo in the trials using OGTT, whereas the trials applying mixed meal tolerance test reported lower postprandial glucose concentration with BAS compared to placebo. While this systematic review with meta-analysis was not designed to investigate the mode of action of BAS, these results could suggest that gallbladder emptying and subsequent release of bile acids into the intestine observed after a mixed meal test as opposed to after an OGTT (Sonne, Rehfeld, Holst, Vilsbøll, & Knop, 2014), is mediating the at least part of the glucose-lowering effect of BASs - potentially via reduced gastric emptying and/or the enteroendocrine hormones glucagon-like peptide-1 (GLP-1) and cholecystokinin (CCK) (Hansen, Sonne, & Knop, 2014).

ACCEPTED MANUSCRIPT

In patients with type 2 diabetes, tight glycemic control may come at the price of weight gain as typically observed when sulfonylurea compounds or insulin treatment are used (Inzucchi et al., 2015). Our results suggest that colesevelam and colestimide are weight neutral, although weight data were only available in a minority of the included trials. However, the relative modest glucose lowering effect of colesevelam and colestimide should be seen in the light of its body weight neutral effect when comparing colesevelam and colestimide to sulfonylurea compounds and insulin treatment.

As aforementioned, the first-generation BAS cholestyramine, in a dose of 24 g daily, has been shown to reduce cardiovascular events in otherwise healthy subjects with hypercholesterolemia (mean LDL cholesterol ~5 mmol/L) not treated with statins ("The Lipid Research Clinics Coronary Primary Prevention Trial results. II," 1984, "The Lipid Research Clinics Coronary Primary Prevention Trial results. I," 1984). The dose may be comparable with the recommended dosing of colesevelam and colestimide, as they are believed to be 4-6 times more potent than cholestyramine (Braunlin W et al., 2000). However, it is difficult to predict the effect of colesevelam and colestimide on cardiovascular events in a population of which a substantial proportion is treated with a statin (Association, 2015). Whether the increase in serum triglycerides associated with BAS treatment is clinical relevant is unknown (Berglund et al., 2012). However, severe hypertriglyceridemia is known to increase the risk of pancreatitis, and may be a risk factor for cardiovascular disease and BAS treatment should be used with caution in patients with high serum triglyceride concentrations and in patients with a history of hypertriglyceridemia-induced pancreatitis (Emerging Risk Factors Collaboration, 2009).

BASs are not absorbed in the intestine and therefore lack systemic toxicity. Also, they are not dependent on liver and kidney function (Lyons et al., 1994). This is in contrast to metformin

ACCEPTED MANUSCRIPT

and sulfonylurea compounds, which should be used with caution in patients with impaired liver and kidney function (Abe, Okada, & Soma, 2011; Marquess, 2011).

In the present analysis, BAS treatment was associated with more serious adverse events than the control group although the absolute number was low. Gastrointestinal adverse events, especially constipation, were prevalent, but did not seem to affect the attrition rate. Poor compliance has been reported with the first-generation BASs cholestyramine and colestipol (Avorn et al., 1998; "The Lipid Research Clinics Coronary Primary Prevention Trial results. II," 1984), but we did not find evidence of differences in attrition rates between patients treated with BASs and the control group.

4.1 Strengths and limitations

This systematic review is the first to include all types of BASs in a systematic review. This provides validity for the study and allows us to make conclusions about the drug class as a whole despite the increased inter-trial heterogeneity. At the same time, by subgroup analysis, we were able to compare the individual BASs. In contrast to a previously published review (Ooi & Loke, 2012), we were able to compile unpublished data from the majority of published trials. These data allowed us to perform rigorous bias control, and to increase the amount of data included in the individual analyses. As outcomes are likely to be presented in a favorable way in published articles, the retrieval of unpublished outcome data may reduce the risk of bias from selective reporting. However, we did not find evidence of publication bias.

Patients lost to follow-up and inadequate randomization can lead to overestimated benefits of a treatment. The majority of the included trials had deficiencies in the reported randomization methods, which is a limitation to the internal validity of our findings. However, all of the

ACCEPTED MANUSCRIPT

included trials but one did account for the intention-to-treat population including losses to follow-up.

The companies marketing colesevelam and colestimide, respectively, funded the vast majority of the included trials. It has been described that financial interests could lead to bias in comparisons of intervention (Wood et al., 2008), but also that funding may be associated with adequate control of bias (Kjaergard & Gluud, 2002). A study of randomized clinical trials showed that industry funding was associated with more positive author conclusions, but not with the quantitative results of the included trials (Als-Nielsen, Chen, Gluud, & Kjaergard, 2003). We did rigorous assessments of the quality of included trials. None of our analyses showed evidence of selection, assessment, attrition, or outcome reporting bias. These analyses support the validity of the results.

4.2 Conclusions

The present systematic review with meta-analyses of randomized controlled trials found that BAS treatment improves glycemic control with a mean reduction in HbA_{1c} of 0.55% (95% CI -0.64 to -0.46) in trials reporting *end of treatment* and 0.40% (95% CI -0.49 to -0.30) in trials reporting *mean change*. Also, treatment with BASs reduces LDL cholesterol, although to a lesser extent than statin treatment, and appears to be weight neutral. BAS treatment is associated with gastrointestinal adverse events, especially constipation. Further, BAS treatment is associated with increased serum triglyceride concentrations, but the clinical implications of this are unknown. Thus, colesevelam and colestimide with their dual - albeit modest - actions on glucose and LDL cholesterol could be relevant as add-on treatment in metformin and statin-treated patients with type 2 diabetes, who do not reach the treatment goals of glycemic control and LDL cholesterol, or in patients who are statin intolerant. Also, BASs could, given the non-absorbable nature, be used in the treatment of patients with type 2

ACCEPTED MANUSCRIPT

diabetes and end-stage renal disease. However, colesevelam and colestimide have not been evaluated in cardiovascular outcome trials, which since 2008 has been required by the FDA for new antidiabetic drugs (FDA, n.d.). Cardiovascular outcome trials are essential to establish the safety and potential of BAS treatment. To our knowledge, no such trials have been planned. Finally, we were only able to identify one trial comparing BAS treatment with other antidiabetic treatments (Rigby et al., 2010). Such trials are also needed to establish the place of BASs in the management of type 2 diabetes.

ACKNOWLEDGEMENTS

We thank Sankyo Daichii for providing additional data from some of the trials funded by the company (Bays et al., 2008; Fonseca et al., 2008; Goldberg et al., 2008; Henry et al., 2012, 2012; Rigby et al., 2010, 2010; J. Rosenstock, Truitt, et al., 2014; J. Rosenstock, Rigby, et al., 2014; Julio Rosenstock et al., 2010; Schwartz et al., 2010). We thank Smushkin *et al*. for providing baseline and outcome data (Smushkin et al., 2013). We thank Mitsubishi Tanabe Pharma Corporation for providing data on the trial published by Kondo *et al.* (Kondo & Kadowaki, 2010). We also thank Takebayashi *et al*. (Takebayashi et al., 2009), Kato et *al.* (Kato et al., 2011), Neda *et al*. (Neda et al., 2012), and Suzuki *et al*. (Suzuki et al., 2012) for providing additional baseline and outcome data.

APPENDIX A. SUPPLEMENTARY DATA

Supplementary data to this article can be found online at: xxxxxx. The protocol is available online at BMJ Open: http://bmjopen.bmj.com/content/2/6/e001803.full?eaf

ACCEPTED MANUSCRIPT

FUNDING AND DISCLOSURES

The present study has not received direct funding. KHM holds shares in Chr. Hansen and Probi. TV has received lecture fees and/or research grants from, participated in advisory boards of and/or consulted for Amgen, AstraZeneca, Boehringer Ingelheim, Bristol-Myers Squibb, Eli Lilly, MSD/Merck, Novo Nordisk and Sanofi. FKK has received lecture fees and/or research grants from, participated in advisory boards of and/or consulted for Amgen, AstraZeneca, Boehringer Ingelheim, Eli Lilly, MSD/Merck, Novo Nordisk, Sanofi and Zealand Pharma. The remaining authors have nothing to disclose. This article does not concern any studies with human or animal subjects performed by the any of the authors. After the initial draft of the manuscript, MH took up a position at Novo Nordisk, Søborg, Denmark.

All authors declare that none have received support from any organization for the present work, no relationship with any organization that may have an interest in the present work, and any other relationships or activities that could appear to have influenced the present work.

Ethical approval: Not required.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

REFERENCES

Abe, M., Okada, K., & Soma, M. (2011). Antidiabetic agents in patients with chronic kidney disease and end-stage renal disease on dialysis: metabolism and clinical practice. *Current Drug Metabolism*, *12*(1), 57–69.

ADVANCE Collaborative Group, Patel, A., MacMahon, S., Chalmers, J., Neal, B., Billot, L., … Travert, F. (2008). Intensive blood glucose control and vascular outcomes in patients with type 2 diabetes. *The New England Journal of Medicine*, *358*(24), 2560–2572. https://doi.org/10.1056/NEJMoa0802987

Als-Nielsen, B., Chen, W., Gluud, C., & Kjaergard, L. L. (2003). Association of funding and conclusions in randomized drug trials: a reflection of treatment effect or adverse events? *JAMA: The Journal of the American Medical Association*, *290*(7), 921–928. https://doi.org/10.1001/jama.290.7.921

Association, A. D. (2015). 8. Cardiovascular Disease and Risk Management. *Diabetes Care*, *38*(Supplement 1), S49–S57. https://doi.org/10.2337/dc15-S011

Avorn, J., Monette, J., Lacour, A., Bohn, R. L., Monane, M., Mogun, H., & LeLorier, J. (1998). Persistence of use of lipid-lowering medications: a cross-national study. *JAMA: The Journal of the American Medical Association*, *279*(18), 1458–1462.

Bays, H. E., Goldberg, R. B., Truitt, K. E., & Jones, M. R. (2008). Colesevelam hydrochloride therapy in patients with type 2 diabetes mellitus treated with metformin: glucose and lipid effects. *Archives of Internal Medicine*, *168*(18), 1975–1983. https://doi.org/10.1001/archinte.168.18.1975

Berglund, L., Brunzell, J. D., Goldberg, A. C., Goldberg, I. J., Sacks, F., Murad, M. H., … Endocrine society. (2012). Evaluation and treatment of hypertriglyceridemia: an Endocrine Society clinical practice guideline. *The Journal of Clinical Endocrinology and Metabolism*,

ACCEPTED MANUSCRIPT

97(9), 2969–2989. https://doi.org/10.1210/jc.2011-3213

Beysen, C., Murphy, E. J., Deines, K., Chan, M., Tsang, E., Glass, A., … Hellerstein, M. K. (2012). Effect of bile acid sequestrants on glucose metabolism, hepatic de novo lipogenesis, and cholesterol and bile acid kinetics in type 2 diabetes: a randomised controlled study. *Diabetologia*, *55*(2), 432–442. https://doi.org/10.1007/s00125-011-2382-3

Braunlin W, Zhorov E, Smisek D, Guo A, Appruzese W, Xu Q, … Mandeville H. (2000). In vitro comparison of bile acid binding to colesevelam HCl and other bile acid sequestrants. *Polymer Preprints*, *41*, 708– 709.

Brufau, G., Stellaard, F., Prado, K., Bloks, V. W., Jonkers, E., Boverhof, R., … Murphy, E. J. (2010). Improved glycemic control with colesevelam treatment in patients with type 2 diabetes is not directly associated with changes in bile acid metabolism. *Hepatology (Baltimore, Md.)*, *52*(4), 1455–1464. https://doi.org/10.1002/hep.23831

Cheung, B. M. Y., Ong, K. L., Cherny, S. S., Sham, P.-C., Tso, A. W. K., & Lam, K. S. L. (2009). Diabetes prevalence and therapeutic target achievement in the United States, 1999 to 2006. *The American Journal of Medicine*, *122*(5), 443–453. https://doi.org/10.1016/j.amjmed.2008.09.047

Cochrane Handbook for Systematic Reviews of Interventions | The Cochrane Collaboration. (n.d.). Retrieved May 14, 2012, from http://handbook.cochrane.org/

Commissioner, O. of the. (n.d.). Drug Safety Labeling Changes - WelChol (colesevelam hydrochloride) Tablets [WebContent]. Retrieved October 8, 2012, from http://www.fda.gov/Safety/MedWatch/SafetyInformation/Safety-RelatedDrugLabelingChanges/ucm106676.htm

Emerging Risk Factors Collaboration, Di Angelantonio, E., Sarwar, N., Perry, P., Kaptoge, S., Ray, K. K., … Danesh, J. (2009). Major lipids, apolipoproteins, and risk of vascular disease.

ACCEPTED MANUSCRIPT

JAMA: The Journal of the American Medical Association, *302*(18), 1993–2000. https://doi.org/10.1001/jama.2009.1619

FDA. (n.d.). Guidance for Industry Diabetes Mellitus - Evaluating Cardiovascular Risk in New Antidiabetic Therapies to Treat Type 2 Diabetes. Retrieved from http://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/ucm071627.pdf

Fonseca, V. A., Rosenstock, J., Wang, A. C., Truitt, K. E., & Jones, M. R. (2008). Colesevelam HCl improves glycemic control and reduces LDL cholesterol in patients with inadequately controlled type 2 diabetes on sulfonylurea-based therapy. *Diabetes Care*, *31*(8), 1479–1484. https://doi.org/10.2337/dc08-0283

Gaede, P., Vedel, P., Larsen, N., Jensen, G. V. H., Parving, H.-H., & Pedersen, O. (2003). Multifactorial intervention and cardiovascular disease in patients with type 2 diabetes. *The New England Journal of Medicine*, *348*(5), 383–393. https://doi.org/10.1056/NEJMoa021778

Gall, M. A., Borch-Johnsen, K., Hougaard, P., Nielsen, F. S., & Parving, H. H. (1995). Albuminuria and poor glycemic control predict mortality in NIDDM. *Diabetes*, *44*(11), 1303–1309.

Garber, A. J., Abrahamson, M. J., Barzilay, J. I., Blonde, L., Bloomgarden, Z. T., Bush, M. A., … Davidson, M. H. (2013). AACE comprehensive diabetes management algorithm 2013. *Endocrine Practice: Official Journal of the American College of Endocrinology and the American Association of Clinical Endocrinologists*, *19*(2), 327–336.

Garg, A., & Grundy, S. M. (1994). Cholestyramine therapy for dyslipidemia in non-insulin-dependent diabetes mellitus. A short-term, double-blind, crossover trial. *Annals of Internal Medicine*, *121*(6), 416–422.

Goldberg, R. B., Fonseca, V. A., Truitt, K. E., & Jones, M. R. (2008). Efficacy and safety of

ACCEPTED MANUSCRIPT

colesevelam in patients with type 2 diabetes mellitus and inadequate glycemic control receiving insulin-based therapy. *Archives of Internal Medicine*, *168*(14), 1531–1540. https://doi.org/10.1001/archinte.168.14.1531

Hansen, M., Sonne, D. P., & Knop, F. K. (2014). Bile acid sequestrants: glucose-lowering mechanisms and efficacy in type 2 diabetes. *Current Diabetes Reports*, *14*(5), 482. https://doi.org/10.1007/s11892-014-0482-4

Hansen, M., Sonne, D. P., Mikkelsen, K. H., Gluud, L. L., Vilsbøll, T., & Knop, F. K. (2012). Effect of bile acid sequestrants on glycaemic control: protocol for a systematic review with meta-analysis of randomised controlled trials. *BMJ Open*, *2*(6). https://doi.org/10.1136/bmjopen-2012-001803

Henry, R. R., Aroda, V. R., Mudaliar, S., Garvey, W. T., Chou, H. S., & Jones, M. R. (2012). Effects of colesevelam on glucose absorption and hepatic/peripheral insulin sensitivity in patients with type 2 diabetes mellitus. *Diabetes, Obesity & Metabolism*, *14*(1), 40–46. https://doi.org/10.1111/j.1463-1326.2011.01486.x

Higgins, J. P. T., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, *21*(11), 1539–1558. https://doi.org/10.1002/sim.1186

Inzucchi, S. E., Bergenstal, R. M., Buse, J. B., Diamant, M., Ferrannini, E., Nauck, M., … Matthews, D. R. (2015). Management of Hyperglycemia in Type 2 Diabetes, 2015: A Patient-Centered Approach: Update to a Position Statement of the American Diabetes Association and the European Association for the Study of Diabetes. *Diabetes Care*, *38*(1), 140–149. https://doi.org/10.2337/dc14-2441

Kato, T., Inagaki, K., Sawai, Y., Kanayama, H., Katada, N., & Itoh, M. (2011). Comparison of efficacy of pitavastatin and colestimide in Japanese patients with diabetes mellitus complicated by hyperlipidemia and metabolic syndrome. *Experimental and Clinical*

ACCEPTED MANUSCRIPT

Endocrinology & Diabetes: Official Journal, German Society of Endocrinology [and] German Diabetes Association, *119*(9), 554–558. https://doi.org/10.1055/s-0031-1273770

Kjaergard, L. L., & Gluud, C. (2002). Funding, disease area, and internal validity of hepatobiliary randomized clinical trials. *The American Journal of Gastroenterology*, *97*(11), 2708–2713. https://doi.org/10.1111/j.1572-0241.2002.07067.x

Kondo, K., & Kadowaki, T. (2010). Colestilan monotherapy significantly improves glycaemic control and LDL cholesterol levels in patients with type 2 diabetes: a randomized double-blind placebo-controlled study. *Diabetes, Obesity & Metabolism*, *12*(3), 246–251. https://doi.org/10.1111/j.1463-1326.2009.01159.x

Lyons, D., Webster, J., Fowler, G., & Petrie, J. C. (1994). Colestipol at varying dosage intervals in the treatment of moderate hypercholesterolaemia. *British Journal of Clinical Pharmacology*, *37*(1), 59–62.

Marina, A. L., Utzschneider, K. M., Wright, L. A., Montgomery, B. K., Marcovina, S. M., & Kahn, S. E. (2012). Colesevelam improves oral but not intravenous glucose tolerance by a mechanism independent of insulin sensitivity and β-cell function. *Diabetes Care*, *35*(5), 1119–1125. https://doi.org/10.2337/dc11-2050

Marquess, J. G. (2011). Managing special populations among patients with type 2 diabetes mellitus. *Pharmacotherapy*, *31*(12 Suppl), 65S–72S. https://doi.org/10.1592/phco.31.12.65S

Mazze, R. S., Strock, E. S., Monk, A. M., Murphy, M. M., Xi, M., & Bergenstal, R. M. (2013). Diurnal glucose profiles using continuous glucose monitoring to identify the glucose-lowering characteristics of colesevelam HCl (Welchol). *Endocrine Practice: Official Journal of the American College of Endocrinology and the American Association of Clinical Endocrinologists*, *19*(2), 275–283. https://doi.org/10.4158/EP12337.OR

Moher, D., Liberati, A., Tetzlaff, J., & Altman, D. G. (2009). Preferred reporting items for

ACCEPTED MANUSCRIPT

systematic reviews and meta-analyses: the PRISMA statement. *BMJ (Clinical Research Ed.)*, *339*, b2535.

Neda, T., Inukai, K., Kurihara, S., Ono, H., Hosaka, T., Nakamoto, H., … Awata, T. (2012). Hypoglycemic effects of colestimide on type 2 diabetic patients with obesity. *Endocrine Journal*, *59*(3), 239–246.

Nissen, S. E., & Wolski, K. (2010). Rosiglitazone revisited: an updated meta-analysis of risk for myocardial infarction and cardiovascular mortality. *Archives of Internal Medicine*, *170*(14), 1191–1201. https://doi.org/10.1001/archinternmed.2010.207

Ooi, C. P., & Loke, S. C. (2012). Colesevelam for type 2 diabetes mellitus. *The Cochrane Database of Systematic Reviews*, *12*, CD009361. https://doi.org/10.1002/14651858.CD009361.pub2

Rigby, S. P., Handelsman, Y., Lai, Y.-L., Abby, S. L., Tao, B., & Jones, M. R. (2010). Effects of colesevelam, rosiglitazone, or sitagliptin on glycemic control and lipid profile in patients with type 2 diabetes mellitus inadequately controlled by metformin monotherapy. *Endocrine Practice: Official Journal of the American College of Endocrinology and the American Association of Clinical Endocrinologists*, *16*(1), 53–63. https://doi.org/10.4158/EP09146.OR

Rosenbaum, D. P., Petersen, J. S., Ducharme, S., Markham, P., & Goldberg, D. I. (1997). Absorption, distribution and excretion of GT31-104, a novel bile acid sequestrant, in rats and dogs after acute and subchronic administration. *Journal of Pharmaceutical Sciences*, *86*(5), 591–595. https://doi.org/10.1021/js9603820

Rosenstock, J., Fonseca, V. A., Garvey, W. T., Goldberg, R. B., Handelsman, Y., Abby, S. L., … Jones, M. R. (2010). Initial combination therapy with metformin and colesevelam for achievement of glycemic and lipid goals in early type 2 diabetes. *Endocrine Practice: Official Journal of the American College of Endocrinology and the American Association of Clinical*

ACCEPTED MANUSCRIPT

Endocrinologists, *16*(4), 629–640. https://doi.org/10.4158/EP10130.OR

Rosenstock, J., Rigby, S. P., Ford, D. M., Tao, B., & Chou, H. S. (2014). The glucose and lipid effects of colesevelam as monotherapy in drug-naïve type 2 diabetes. *Hormone and Metabolic Research = Hormon- Und Stoffwechselforschung = Hormones Et Métabolisme*, *46*(5), 348–353. https://doi.org/10.1055/s-0033-1358759

Rosenstock, J., Truitt, K. E., Baz-Hecht, M., Ford, D. M., Tao, B., & Chou, H. S. (2014). Efficacy and Safety of Colesevelam in Combination with Pioglitazone in Patients with Type 2 Diabetes Mellitus. *Hormone and Metabolic Research = Hormon- Und Stoffwechselforschung = Hormones Et Metabolisme*. https://doi.org/10.1055/s-0034-1383648

Schwartz, S. L., Lai, Y.-L., Xu, J., Abby, S. L., Misir, S., Jones, M. R., & Nagendran, S. (2010). The effect of colesevelam hydrochloride on insulin sensitivity and secretion in patients with type 2 diabetes: a pilot study. *Metabolic Syndrome and Related Disorders*, *8*(2), 179–188. https://doi.org/10.1089/met.2009.0049

Smushkin, G., Sathananthan, M., Piccinini, F., Dalla Man, C., Law, J. H., Cobelli, C., … Vella, A. (2013). The effect of a bile acid sequestrant on glucose metabolism in subjects with type 2 diabetes. *Diabetes*, *62*(4), 1094–1101. https://doi.org/10.2337/db12-0923

Sonne, D. P., Rehfeld, J. F., Holst, J. J., Vilsbøll, T., & Knop, F. K. (2014). Postprandial gallbladder emptying in patients with type 2 diabetes: potential implications for bile-induced secretion of glucagon-like peptide 1. *European Journal of Endocrinology / European Federation of Endocrine Societies*, *171*(4), 407–419. https://doi.org/10.1530/EJE-14-0309

Stratton, I. M., Adler, A. I., Neil, H. A., Matthews, D. R., Manley, S. E., Cull, C. A., … Holman, R. R. (2000). Association of glycaemia with macrovascular and microvascular complications of type 2 diabetes (UKPDS 35): prospective observational study. *BMJ (Clinical Research Ed.)*, *321*(7258), 405–412.

ACCEPTED MANUSCRIPT

Suzuki, T., Oba, K., Igari, Y., Watanabe, K., Matsumura, N., Futami-Suda, S., … Nakano, H. (2012). Effects of bile-acid-binding resin (colestimide) on blood glucose and visceral fat in Japanese patients with type 2 diabetes mellitus and hypercholesterolemia: an open-label, randomized, case-control, crossover study. *Journal of Diabetes and Its Complications*, *26*(1), 34–39. https://doi.org/10.1016/j.jdiacomp.2011.11.008

Takebayashi, K., Suetsugu, M., Matsumoto, S., Aso, Y., & Inukai, T. (2009). Effects of rosuvastatin and colestimide on metabolic parameters and urinary monocyte chemoattractant protein-1 in type 2 diabetic patients with hyperlipidemia. *Southern Medical Journal*, *102*(4), 361–368. https://doi.org/10.1097/SMJ.0b013e31819bd023

The Lipid Research Clinics Coronary Primary Prevention Trial results. II. The relationship of reduction in incidence of coronary heart disease to cholesterol lowering. (1984). *JAMA: The Journal of the American Medical Association*, *251*(3), 365–374.

The Lipid Research Clinics Coronary Primary Prevention Trial results. I. Reduction in incidence of coronary heart disease. (1984). *JAMA*, *251*(3), 351–364.

Wood, L., Egger, M., Gluud, L. L., Schulz, K. F., Jüni, P., Altman, D. G., … Sterne, J. A. C. (2008). Empirical evidence of bias in treatment effect estimates in controlled trials with different interventions and outcomes: meta-epidemiological study. *BMJ (Clinical Research Ed.)*, *336*(7644), 601–605. https://doi.org/10.1136/bmj.39465.451748.AD

Yamakawa, T., Takano, T., Utsunomiya, H., Kadonosono, K., & Okamura, A. (2007). Effect of colestimide therapy for glycemic control in type 2 diabetes mellitus with hypercholesterolemia. *Endocrine Journal*, *54*(1), 53–58.

Zieve, F. J., Kalin, M. F., Schwartz, S. L., Jones, M. R., & Bailey, W. L. (2007). Results of the glucose-lowering effect of WelChol study (GLOWS): a randomized, double-blind, placebo-controlled pilot study evaluating the effect of colesevelam hydrochloride on glycemic

ACCEPTED MANUSCRIPT

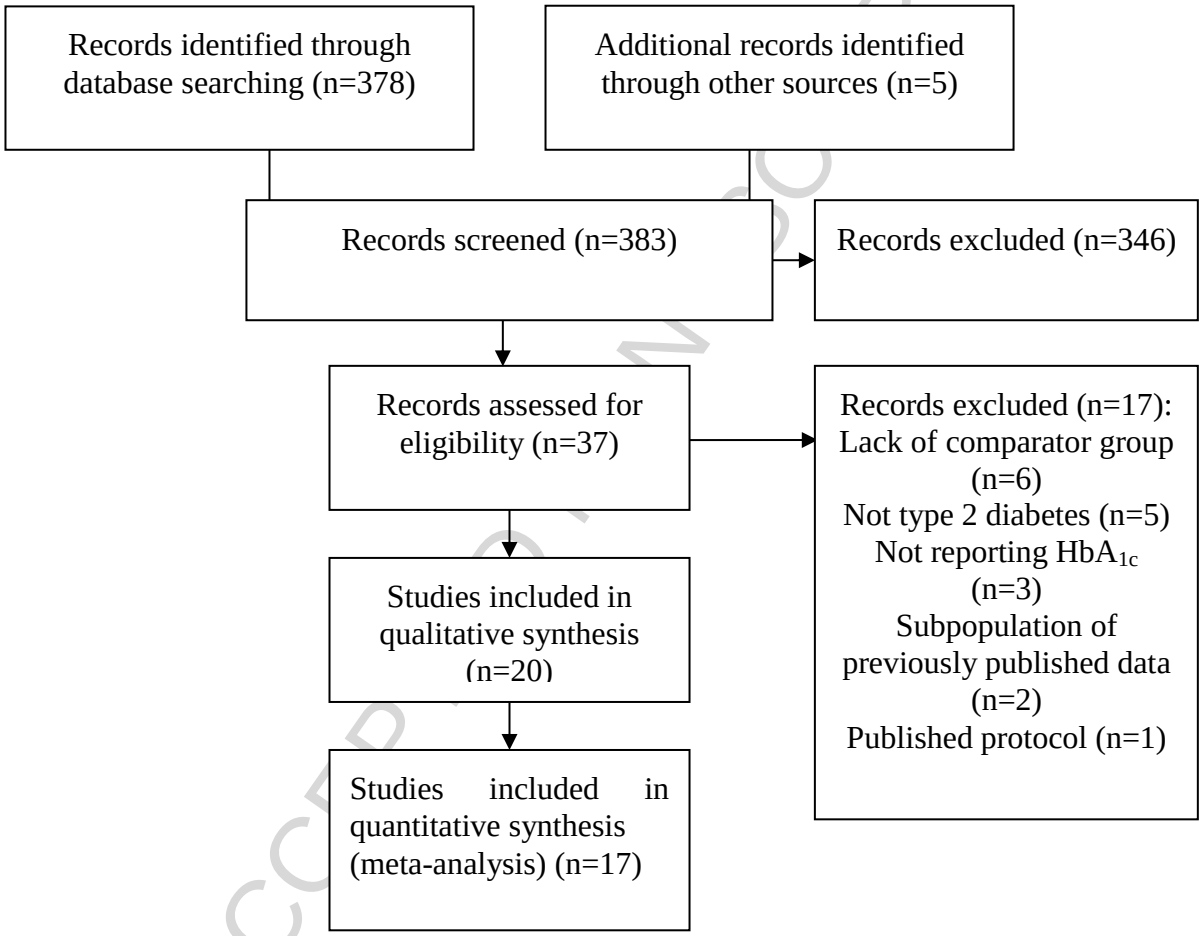
control in subjects with type 2 diabetes. *Clinical Therapeutics*, *29*(1), 74–83. https://doi.org/10.1016/j.clinthera.2007.01.003

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

FIGURES

Fig 1. Identification and inclusion of randomized controlled trials



ACCEPTED MANUSCRIPT

Fig 2.A *End of treatment* HbA_{1c} random-effects meta-analysis

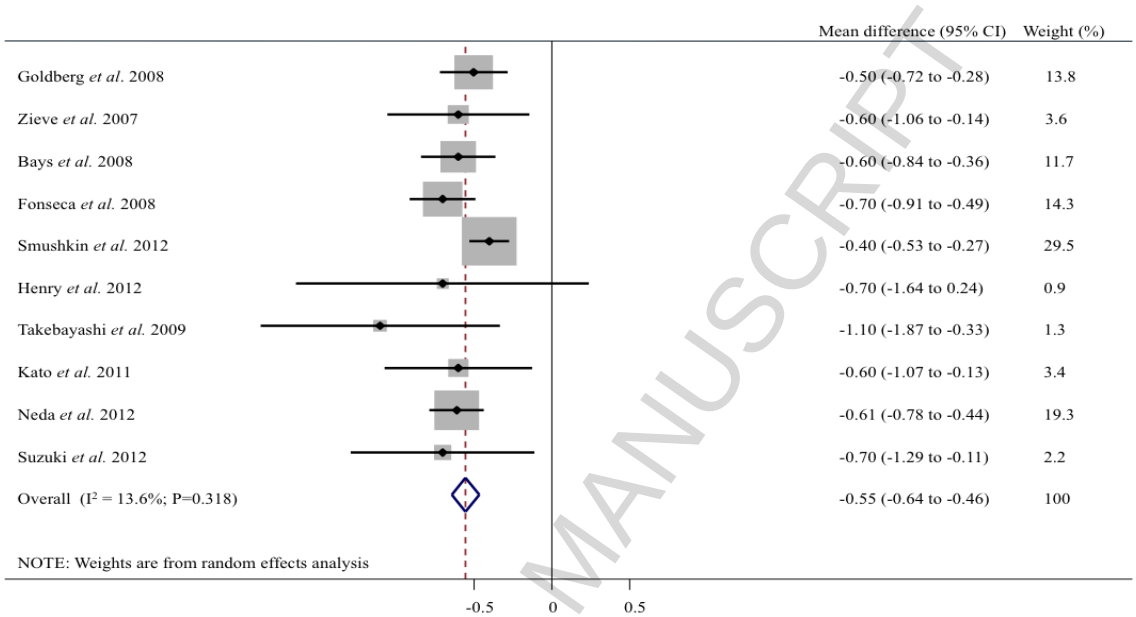


Fig 2.B *End of trial* fasting plasma glucose random-effects meta-analysis

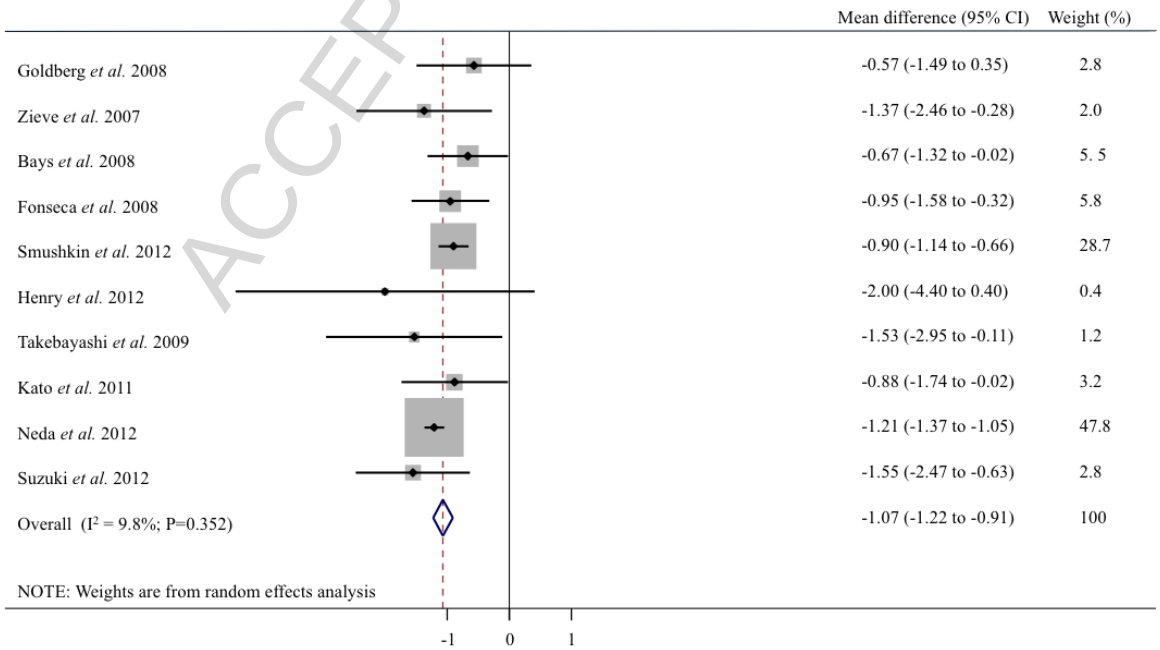


Fig 2. A. Random-effects meta-analysis of trials reporting *end of treatment* HbA_{1c} (%). **B.** Random-effects meta-analysis of trials reporting *end of treatment* fasting plasma glucose (mmol/L).

ACCEPTED MANUSCRIPT

TABLES

Table 1. Characteristics of included trials

Trial	Intervention	Location	Concomitant antidiabetic treatment	Duration (weeks)	Primary endpoint
Zieve *et al*. 2007	Colesevelam (3.75 g/d) versus placebo	15 sites; USA	Metformin; SU	12	HbA_{1c} mean change
Bays *et al*. 2008	Colesevelam (3.75 g/d) versus placebo	56 sites; USA, Mexico	Metformin; metformin combined with another OAD	26	HbA_{1c} mean change
Fonseca *et al*. 2008	Colesevelam(3.75 g/d) versus placebo	51 sites; USA, Mexico	SU; SU combined with another OAD	26	HbA_{1c} mean change
Goldberg *et al*. 2008	Colesevelam (3.75 g/d) versus placebo	51 sites; USA, Mexico	Insulin; insulin combined with an OAD	16	HbA_{1c} mean change
Rigby *et al*. 2010	Colesevelam (3.75 g/d) versus sitagliptin (100 mg/d)	33 sites; USA, Mexico, Colombia	Metformin	16	HbA_{1c} mean change
Rosenstock *et al*. 2010	Colesevelam (3.75 g/d) versus placebo	38 sites; USA, Mexico, Colombia, India	Metformin (initiated at inclusion)	16	HbA_{1c} mean change
Schwartz *et al*. 2010	Colesevelam (3.75 g/d) versus placebo	1 site; USA	OAD were withdrawn 2 weeks prior to inclusion	8	Glucose-lowering mechanism
Beysen *et al*. 2012	Colesevelam (3.75 g/d) versus placebo	3 sites; USA	Meglitinide; metformin; metformin-SU; SU; TZD; insulin	12	Glucose-lowering mechanism
Henry *et al*. 2012	Colesevelam (3.75 g/d) versus placebo	2 sites; USA	SU (all non-SU OADs were withdrawn up to 18 weeks prior to inclusion)	12	Glucose-lowering mechanism
Smushkin *et al*. 2012	Colesevelam (3.75 g/d) versus placebo	1 site; USA	Metformin	12	Glucose-lowering mechanism
Rosenstock *et al*. 2013	Colesevelam (3.75 g/d) versus placebo	74 sites; USA	None	24	HbA_{1c} mean change
Rosenstock *et al*. 2014	Colesevelam (3.75 g/d) versus placebo	145 sites; USA	Pioglitazone; pioglitazone combined with another OAD	24	HbA_{1c} mean change
Takebayashi *et al*. 2009	Colestimide (3.0 g/d) versus rosuvastatin	1 site; Japan	Diet; metformin; alpha-GI; meglitinide; SU; TZD; insulin	12	Glucose-lowering efficacy
Kondo *et al*. 2010	Colestimide (4.5 g/d) versus placebo	44 sites; Japan	Diet; alpha-GI; SU; insulin	12	Glucose-lowering efficacy
Kato *et al*. 2011	Colestimide (3.0 g/d) versus pitavastatin	1 site; Japan	Diet; alpha-GI; metformin; SU; TZD	24	Glucose-lowering efficacy
Neda et al. 2012	Colestimide (4.0 g/d) versus no intervention	1 site; Japan	Unspecified OADs	24	Glucose-lowering efficacy
Suzuki *et al*. 2012	Colestimide (3.0 g/d) versus no intervention	1 site; Japan	Diet; alpha-GI; meglitinide; metformin; SU; TZD	12	Glucose-lowering mechanism

Interventions are shown as gram per day. Alpha-GI, alpha glucosidase inhibitor; OAD, oral anti-diabetic drug; SU, sulfonylurea; TZD, thiazolidinedione. The trials are listed alphabetically by intervention and by year of publication.

ACCEPTED MANUSCRIPT

Table 2. Baseline characteristics of patients with type 2 diabetes

Trial	No of patients randomised	Age (years)*	BMI (Kg/m^2)*	Duration of diabetes (years)*	HbA_{1c} (%)*
Zieve *et al*. 2007	31/34	56.7 (9.1)/55.7 (9.7)	32.5 (5.2)/32.2 (5.5)	NR	7.9 (0.8)/8.1 (0.9)
Bays *et al*. 2008	159/157	55.7 (9.6)/56.9 (9.5)	33.9 (5.3)/33.0 (6.3)	NR	8.2 (0.7)/8.1 (0.6)
Fonseca *et al*. 2008	230/231	56.6 (10.3)/57.0 (10.3)	33.1 (5.95)/32.5 (5.64)	NR	8.2 (0.68)/8.3 (0.72)
Goldberg *et al*. 2008	147/140	56.9 (9.8)/56.3 (9.3)	34.9 (5.82)/34.9(5.91)	NR	8.3 (0.62)/8.3 (0.63)
Rigby *et al*. 2010	57/56	56.6 (9.2)/54.8 (9.8)	NR	8.1 (5.9)/8.4 (7.1)	8.1 (0.77)/8.17 (0.91)
Rosenstock *et al*. 2010	145/141	52.7 (11.5)/53.9 (10.1)	30.6 (4.7)/29.8 (4.4)	NR	7.8 (1.0)/7.5 (0.9)
Schwartz *et al*. 2010	17/18	51.4 (12.7)/56.0 (7.9)	35.2 (3.6)/33.4 (5.2)	NR	8.2 (0.9)/8.7 (0.9)
Beysen *et al*. 2012	30/30	59 (9)/56 (9)	30 (5)/31 (5)	NR	8.5 (1.2)/8.0 (0.9)
Henry *et al*. 2012	16/14	55.7 (9.6)/54.0 (10.3)	33.7 (6.77)/33.7 (5.78)	NR	8.2 (0.82)/8.5 (0.81)
Smushkin *et al*. 2012	19/19	62.6 (5.7)/60.2 (6.5)	30.8 (4.4)/30.4 (3.9)	NR	6.7 (0.4)/ 6.8 (0.4)
Rosenstock *et al*. 2013	176/181	52.6 (10.3)/51.8 (10.5)	32.0 (6.5)/31.8 (4.9)	4.3 (4.7)/3.9 (4.4)	8.3 (0.7)/8.2 (0.7)
Rosenstock *et al*. 2014	280/282	56.3 (10.2)/55.6 (10.6)	31.9 (4.8)/31.7 (4.7)	8.5 (6.4)/8.3 (7.0)	8.2 (0.7)/8.1 (0.7)
Takebayashi *et al*. 2009	20/20	62.7 (10.4)/56.2 (14.9)	24.5 (4.2)/25.6 (4.4)	14.7 (7.1)/13.0 (8.1)	7.9 (1.3)/8.5 (1.7)
Kondo *et al*. 2010	92/91	57.7 (11.0)/59.6 (10.1)	26.1 (4.3)/25.9 (4.2)	4.6 (4.6)/5.2 (4.8)	8.0 (0.7)/7.9 (0.8)
Kato *et al*. 2011	24/24	55.5 (14.6)/63.2 (16.4)	25.7 (2.5)/25.6 (2.5)	4.6 (1.7)/5 (1.9)	6.9 (1)/6.8 (0.8)
Neda *et al*. 2012	23/20	51.1 (2.0)/52.5 (1.9)	30.6 (1.2)/29.0 (0.7)	NR	7.71 (0.32)/7.53 (0.39)
Suzuki *et al*. 2012	20/21	67.2 (9.2)/67.1 (10.7)	25.5 (2.7)/24.8 (3.8)	NR	7.4 (0.9)/7.3 (0.8)

Data are shown as intervention group (colesevelam or colestimide)/control group (placebo; sitagliptin; pravastatin; rosuvastatin; pitavastatin; no intervention). *Data for continuous variables are mean with standard deviation in brackets. BMI, body mass index; HbA_{1c}, hemoglobin A_{1c}; NR, not reported. The trials are listed alphabetically by intervention and by year of publication.

ACCEPTED MANUSCRIPT

Table 3. Risk of bias assessment

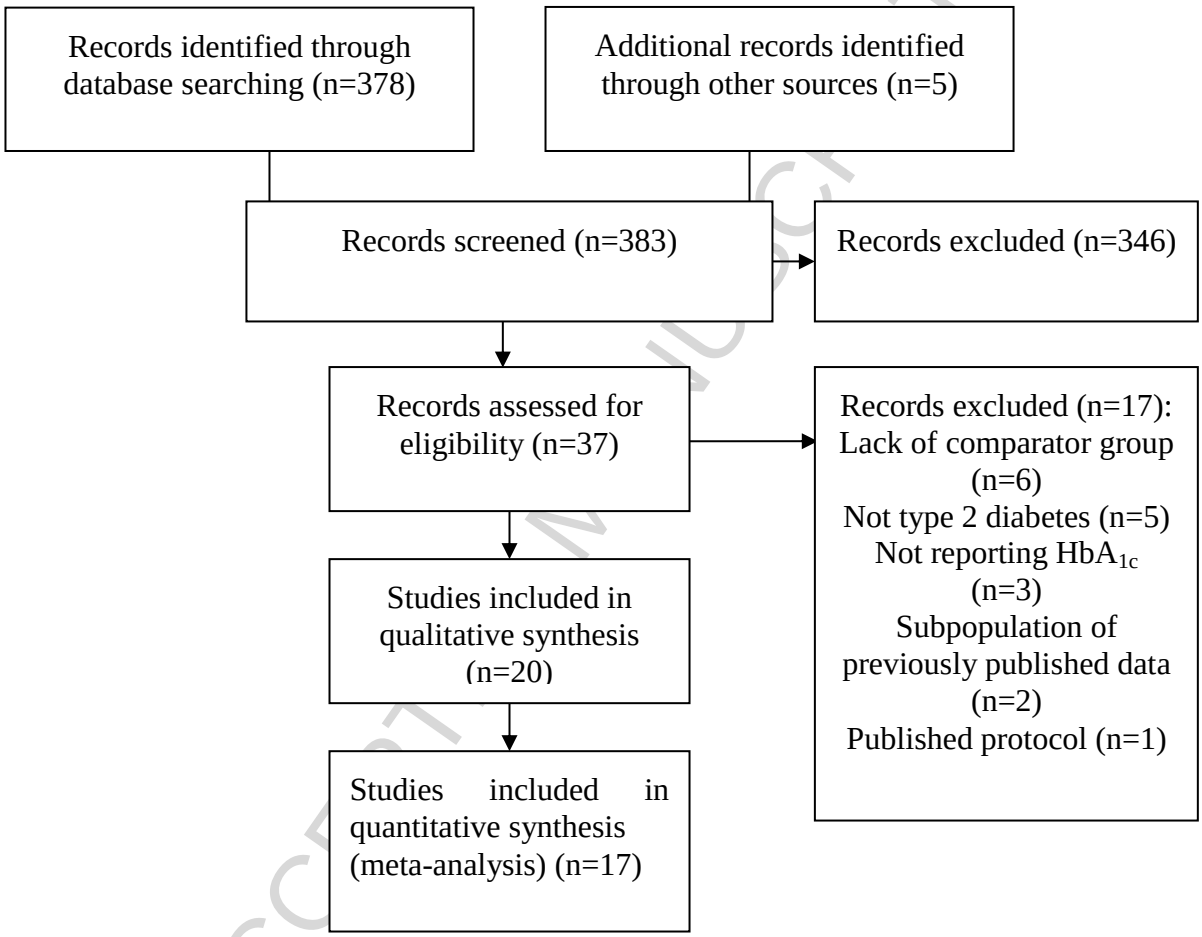
Trial	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of patients, personnel and outcome assessment (performance bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias	Unpublished data retrieved
Zieve *et al*. 2007	Low risk	Low risk	Low risk	Low risk	Uncertain risk	No	Sequence generation and concealment
Bays *et al*. 2008	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment
Fonseca *et al*. 2008	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment, and blood lipids
Goldberg *et al*. 2008	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment, and blood lipids
Rigby *et al*. 2010	Low risk	Low risk	High risk	Low risk	Low risk	No	Sequence generation and concealment, blood lipids and body weight
Rosenstock *et al*. 2010	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment
Schwartz *et al*. 2010	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment
Beysen *et al*. 2012	Low risk	Low risk	Low risk	Low risk	Low risk	No	
Henry *et al*. 2012	Low risk	Low risk	Low risk	Low risk	Low risk	No	HbA_{1c}, FPG, body weight
Smushkin *et al*. 2012	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment, HbA_{1c} and FPG
Rosenstock *et al*. 2013	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment, HbA_{1c} and FPG
Rosenstock *et al*. 2014	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment, HbA_{1c} and FPG
Takebayashi *et al*. 2009	Low risk	Low risk	High risk	Low risk	Low risk	No	Sequence generation and concealment, HbA_{1c} and FPG
Kondo *et al*. 2010	Low risk	Low risk	Low risk	Low risk	Low risk	No	Sequence generation and concealment, HbA_{1c} and FPG
Kato et al. 2011	Uncertain risk	Low risk	High risk	Low risk	Low risk	No	HbA_{1c} and FPG
Neda et al. 2012	Low risk	Low risk	High risk	Low risk	Low risk	No	Sequence generation and concealment, HbA_{1c} and FPG
Suzuki *et al*. 2012	Uncertain risk	Uncertain risk	High risk	Low risk	Low risk	No	HbA_{1c} and FPG, and blood lipids

Risk of bias summary for each bias domain for each of the included trials as assessed by the review authors. The trials are listed alphabetically by intervention and by year of publication.

ACCEPTED MANUSCRIPT

FIGURES

Fig 1. Identification and inclusion of randomised controlled trials



ACCEPTED MANUSCRIPT

Fig 2A *End of treatment* HbA_{1c} random-effects meta-analysis

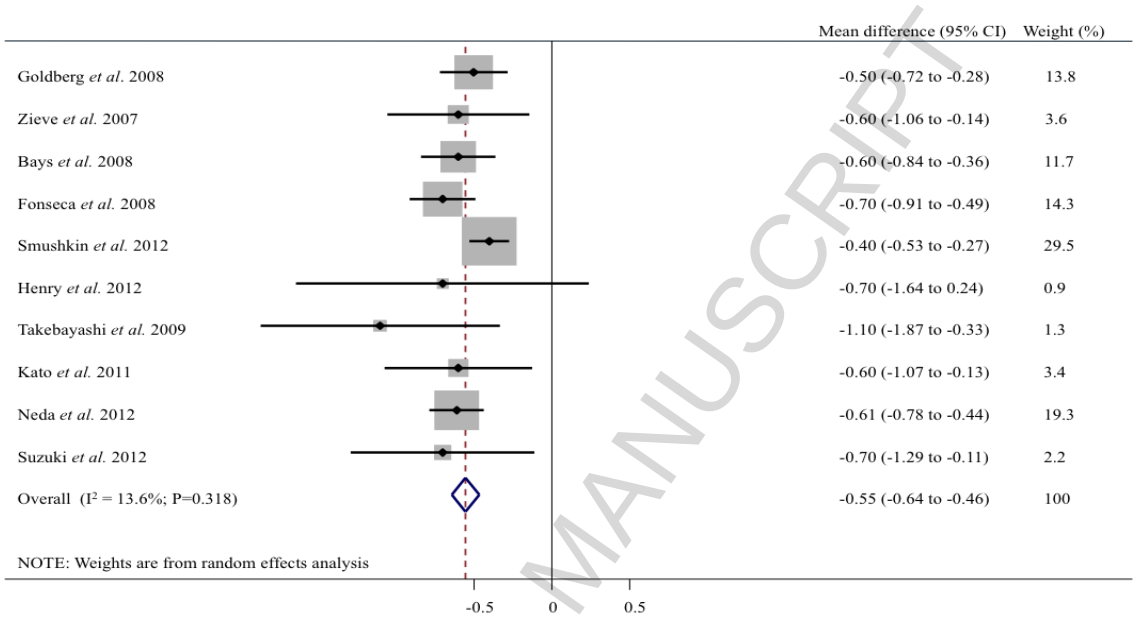
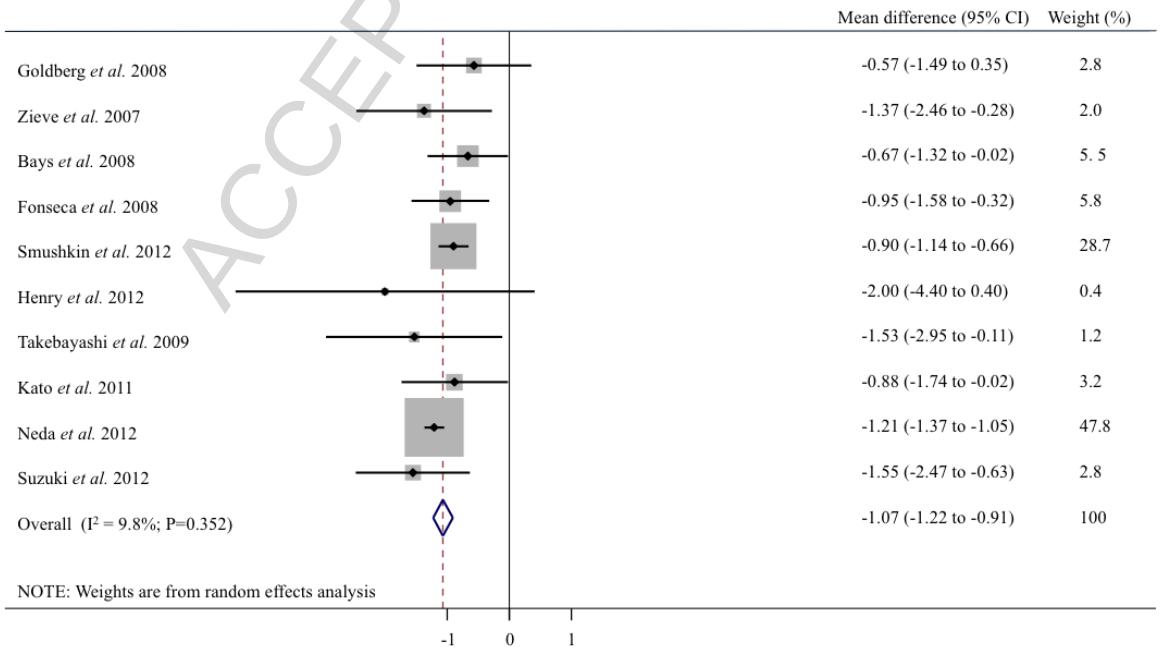
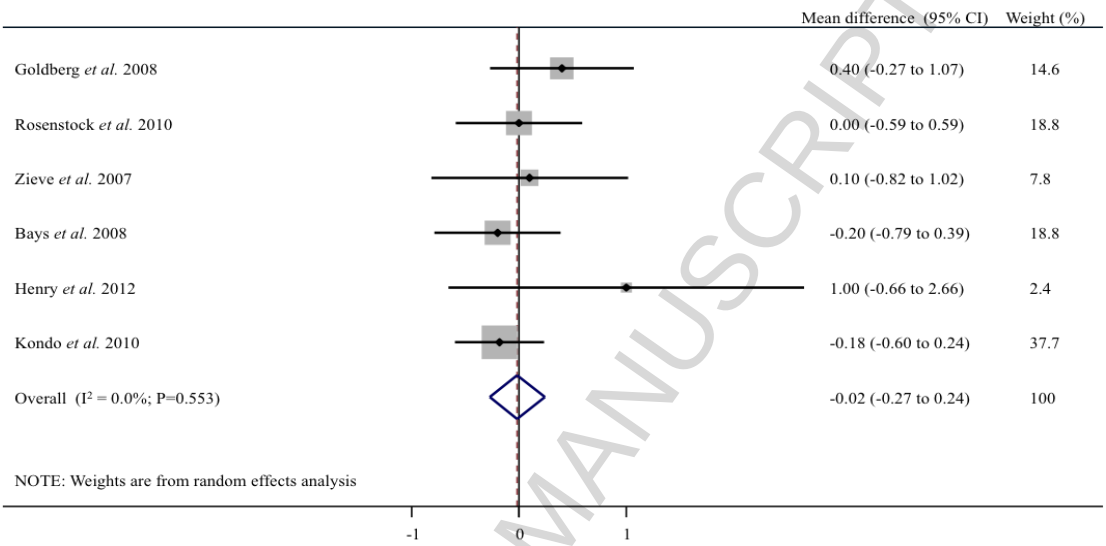


Fig 2B. *End of trial* fasting plasma glucose random-effects meta-analysis



ACCEPTED MANUSCRIPT

Fig 3. ***Mean change*** **in body weight random-effects meta-analysis**



ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

HIGHLIGHTS

- In patients with type 2 diabetes, bile acid sequestrant treatment reduces HbA1c between 0.40% and 0.55%
- Bile acid sequestrants reduce fasting plasma glucose, LDL cholesterol - although to a lesser extent than statin treatment - and appears to be weight neutral
- Bile acid sequestrant treatment is associated with gastrointestinal adverse events especially constipation

ACCEPTED MANUSCRIPT